

CHEMNITZ, Germany

WMO: 105770

Lat: 50.7914N

Lon: 12.8719E

Elev: 417

StdP: 96.41

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.7	-10.8	-17.2	0.9	-11.3	-14.5	1.1	-9.6	17.1	7.5	15.1	4.4	1.9	60	0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	27.9	18.4	25.9	17.6	24.2	16.9	19.4	25.8	18.4	24.3	17.5	22.8	3.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	12.8	21.6	16.2	12.1	20.7	15.3	11.5	19.9	56.8	26.2	53.6	24.4	50.8	23.0	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.9	12.0	10.5	-15.3	31.5	4.7	1.7	-18.7	32.7	-21.4	33.7	-24.1	34.7	-27.5	36.0		
(5)			-15.7	21.2	4.7	0.8	-19.0	21.8	-21.8	22.3	-24.4	22.8	-27.8	23.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-0.3	-0.7	3.5	7.0	12.1	14.8	17.3	17.3	13.6	9.4	4.0	1.3
	DBStd	7.66	5.30	5.49	4.36	4.01	4.03	3.70	3.61	3.63	3.41	4.02	3.94	4.15
	HDD10.0	1478	318	302	204	108	26	3	0	0	5	59	183	271
	HDD18.3	3733	576	534	459	341	196	117	64	64	146	276	431	528
	CDD10.0	871	0	1	3	17	90	147	227	226	114	42	2	1
	CDD18.3	83	0	0	0	0	2	11	33	32	5	1	0	0
	CDH23.3	564	0	0	0	0	11	81	230	218	21	3	0	0
	CDH26.7	114	0	0	0	0	0	12	49	52	1	0	0	0
Wind	WSAvg	4.3	5.5	4.5	4.8	4.0	3.4	3.6	3.2	3.5	4.0	4.4	4.9	5.1
Precipitation	PrecAvg	792	55	48	57	48	70	81	100	87	69	55	63	60
	PrecMax	1013	110	92	153	127	167	186	219	226	129	101	137	108
	PrecMin	515	2	8	14	4	17	41	27	21	25	18	0	23
	PrecStd	117	27	22	32	25	36	33	42	48	31	24	35	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.5	13.7	17.3	20.9	24.7	28.4	30.2	31.0	25.9	22.0	14.9	12.6
		MCWB	7.5	8.9	10.2	12.5	15.9	18.2	19.9	19.4	17.6	15.2	10.6	8.4
	2%	DB	8.3	10.6	13.5	18.1	22.8	25.8	28.1	27.9	23.4	19.2	12.5	10.6
		MCWB	5.8	7.3	9.0	11.2	14.6	17.0	18.4	18.8	16.3	13.5	9.2	7.7
	5%	DB	7.0	8.1	10.8	15.7	21.0	23.5	25.9	25.7	21.4	16.5	10.8	8.7
		MCWB	5.1	5.7	7.1	10.0	13.9	16.4	17.6	17.8	15.5	12.2	8.4	6.5
	10%	DB	5.8	5.6	9.2	13.6	18.9	21.4	24.0	23.8	19.4	14.9	9.2	6.9
		MCWB	4.0	3.5	6.3	8.9	12.8	15.4	16.9	17.0	14.7	11.3	7.1	5.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.9	9.2	11.2	13.0	16.9	19.5	20.7	20.9	18.6	15.9	10.9	9.3
		MCDB	10.2	12.7	15.6	19.2	22.6	25.7	27.8	28.3	23.2	21.6	13.4	11.6
	2%	WB	6.4	7.5	9.0	11.7	15.5	18.2	19.4	19.4	17.2	14.1	9.7	7.9
		MCDB	8.1	10.3	13.2	17.1	21.3	23.5	25.7	26.3	22.1	17.7	11.9	10.2
	5%	WB	5.3	5.4	7.7	10.4	14.5	17.0	18.4	18.3	16.1	12.7	8.5	6.7
		MCDB	6.8	7.9	10.5	15.0	19.7	22.0	24.4	24.4	20.1	15.7	10.6	8.5
	10%	WB	4.1	3.7	6.5	9.2	13.4	16.0	17.4	17.5	15.0	11.7	7.3	5.4
		MCDB	5.6	5.4	9.0	12.9	18.2	20.8	22.9	22.5	18.5	14.3	9.1	6.9
Mean Daily Temperature Range	5% DB	MDBR	4.6	5.3	5.9	7.2	8.2	7.9	8.6	8.5	7.3	6.4	5.1	4.2
		MCDBR	5.9	7.9	8.8	10.5	10.7	11.0	11.6	11.2	10.1	8.8	7.5	5.8
		MCWBR	4.8	5.8	5.8	6.1	5.7	5.4	5.6	5.2	5.3	5.4	5.4	4.5
	5% WB	MCDBR	5.4	7.4	8.3	9.6	9.7	9.7	10.4	10.0	9.0	7.5	6.9	5.3
		MCWBR	4.6	5.8	5.6	5.9	5.8	5.3	5.5	5.1	5.1	5.0	5.3	4.5
Clear-Sky Solar Irradiance	taub	0.298	0.320	0.362	0.401	0.404	0.402	0.417	0.410	0.375	0.349	0.321	0.293	
	taud	2.392	2.358	2.288	2.226	2.251	2.282	2.244	2.282	2.360	2.420	2.437	2.404	
	Ebn at Noon	734	809	832	837	852	855	834	820	811	765	698	681	
	Edh at Noon	62	83	107	127	130	127	130	119	99	78	60	53	
All-Sky Solar Radiation	RadAvg	0.79	1.55	2.52	4.01	4.84	5.22	4.96	4.31	3.08	1.82	0.87	0.61	
	RadStd	0.14	0.24	0.35	0.52	0.59	0.44	0.57	0.40	0.42	0.23	0.11	0.07	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (52 neighbors)		+0.39	N/A	N/A	N/A	+0.31	+0.35	N/A	N/A	+58	+20	

Nomenclature: See separate page